

Enhanced Deep Learning-based Image Dehazing Technique and Comparative Analysis

¹Harish Babu Gade
ECE Department,
CVR College of Engineering
Hyderabad, India
harish.sidhu12@gmail.com

²Venkata Krishna Odugu
ECE Department,
CVR College of Engineering
Hyderabad, India
venkatakrishna.odugu@gmail.com

³B. Janardhana Rao
ECE Department, CVR College of
Engineering Hyderabad, India
janardhan.bitra@gmail.com

⁴Sree Lakshmi Gundebommu
EEE Department,
CVR College of Engineering
Hyderabad, India
g.sreelakshmi@cvr.ac.in

Abstract- Typically, atmospheric variables including dust, smoke, fog, haze, snow, and rain reduce the effectiveness of high-resolution cameras. Degradation can affect the color, contrast, and visual quality of an image. Degraded or hazed images are unsuitable for real-time image processing applications because they lack information specifics. For vision applications, the dehazing step with respect to the images is compulsory to enhance the quality. The primary goal of dehazing is to maintain visual quality while retaining information and scene clarity. An architectural framework for deep learning-based image restoration and single-image dehazing is presented in this work. We propose to reframe the challenge as restoring the image's base component, as opposed to learning a direct mapping between hazy images and their haze-free counterparts, as is the case with most existing approaches. Haze can be removed from a hazy image by first breaking it up into its basic and complex components. Then, by training a CNN to connect the hazy and haze-free fundamental components, additional enhancement of the complicated component can be achieved. The components of the initial image, the haze removed, and the enhanced detail image are combined to generate the final dehazed image.

Keywords: Haze image, Ground truth, CNN, Deep Learning, and Haze Free Image.

I. INTRODUCTION

The image degraded due to haze (or) foggy conditions is called a hazy image or foggy image. Figure 1 displays the contrast between the clear and hazy images. The number of particles in the air is often determined by the surrounding climate and weather. Figure 1 displays the contrast between the clear and hazy images.



Fig. 1. Sample images of without and with Haze.

The unclear images aren't fit for many practical applications like video surveillance, object recognition, remote sensing, transportation surveillance, driver assistance, radar tracking, behavior analysis, etc. To improve the accomplishment of image processing applications concerning these challenging weather variations, image dehazing algorithms are required as a preprocessing tool.

A. Formation of Hazy Image

Haze is an atmospheric parameter that distorts the image. In 1977, The Hazy picture, which is defined by the scattering model of the atmosphere [1] was introduced initially. This model is widely used in many real-time applications. The two main parts of this model are direct attenuation and Airlight as presented in Figure 2.

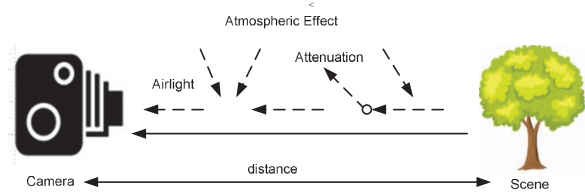


Fig. 2. Effect of attenuation and Airlight.

$$I(x) = J(x)e^{-\beta d(x)} + A(1 - e^{-\beta d(x)}) \quad (1)$$

In the above equation, x provides the input image coordinate information, β is a variable of coefficient of scattering, d is the depth of the scene, J is a haze-free image, and I is a hazy image. The relation between the clear image, also known as the ground truth image, and hazed images is given in Eq. (1).

In Eq. (1), $e^{-\beta d}$ is also depicted as a Transmission Map (TM) is $t(x)$ as provided by Eq. (2).

$$t(x) = e^{-\beta d(x)} \quad (2)$$

Substitute equation (2) is substituted in equation (1), then, the $I(x)$ can be modified as below.

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (3)$$

The value of $\beta \approx 0$, for clear atmosphere, then $I \approx J$. The value of the β is a non-zero and it is not a negligible value. In Eq. (3), $J(x)t(x)$ is referred to as direct attenuation and the $A(1 - t(x))$ is related to Airlight. In the dehazing process, the J is restored from I using the estimation t and A . Then the expression for determination of J using t and A is mentioned in Eq. (4) [2, 18].

$$J(x) = \frac{I(x) - A}{t(x)} + A \quad (4)$$

For obtaining the above J , the estimation of t and A are very important. For the estimation of t and A , proper prior understanding is needed so these values are not estimated simply from I . Here, the t spatially varies with respect to the

d and in the hazy image, the number of unknowns is equal to the pixel number.

II. LITERATURE SURVEY

Dehazing is difficult since haze depends on unknown depth. Moreover, if the only accessible input is a single fuzzy picture, the task becomes more limited and tough. As a result, most classic dehazing techniques have been proposed that rely on numerous hazy photos as input or extra previous knowledge [3], which is generally impracticable in various real-world applications.

An educational model for individual picture dehazing was developed in reference [4] after extensive study of picture attributes associated with haze. In addition, several frameworks for single-image dehazing have been built using Deep Neural Network (DNN). These frameworks take advantage of the rapid advancement of deep learning techniques, which have been very successful in various perceptual tasks such as object detection, image interpretation, and speech recognition. A study suggests that deep CNNs could learn haze-related characteristics automatically, leading to more effective dehazing.

By training on the correlation between hazy images and their related TMs, a DNN with many scales was developed in reference [5] to dehazed individual images. An atmospheric scattering model, also known as MSCNN, may be used to create the dehazed image. The capacity of a trainable end-to-end deep convolutional neural network, also known as DehazeNet, to learn important haze-related characteristics was demonstrated in a prior study where the network was recommended for medium transmission prediction. [7] suggests a CNN-based dehazing network that utilizes a revised atmospheric scattering model to generate dehazed images straight from a single image.

Efficient dehazing methods for both indoor and outdoor images have been suggested in recent studies [17-21]. Various NN approaches are utilized at different phases of the dehazing techniques in this research. Hybrid Meta-Heuristic optimizations are employed in filtering to approximate the atmospheric light of the hazy image. As shown in Figure 3, a distinctive DNN architecture is suggested for single-image dehazing in order to produce more accurate dehazing outcomes with less computational cost. Three key contributions are offered by this research: In order to remove haze, a CNN must be trained to map between hazed & without haze-base components. To counteract the blurring effects seen in many existing approaches, it is necessary to improve the detail component independently.

Our CNN architecture is straightforward and has low run-time complexity, making it appropriate for real-world applications. Finally, our dehazing technique can be repeated multiple times to gradually improve the clarity of output images with no unnecessary improvement.

III. PROPOSED METHOD

The proposed deep learning-based single-image dehazing system comprises three stages: (1) image decomposition-based preprocessing; (2) DNN-based dehazing; and (3) Combining the improved detail image component and the clear component yields the final dehazed image. These three phases are described as follows.

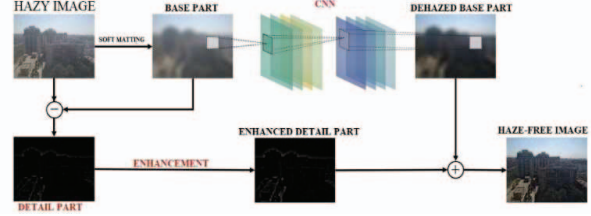


Fig. 3. Important Process steps of the proposed DNN-based dehazing.

A. Preprocessing

Our method involves utilizing image filtering to divide a hazy input image into its base component and detail component to avoid potential loss of details during the dehazing process. The idea behind this is to store low-frequency information like haze in the base component and fine details like textures and edges in the detail component. The widely used soft matting image filter [8], with its effective edge-preserving flattening properties in linear time, is used to accomplish this purpose. We use soft matting filtering to the hazy input image (I) to separate it into base and detail components as described in Eq. (5).

$$I = I_{base\ part} + I_{detail\ Part} \quad (5)$$

Moreover, several other applications have demonstrated the effectiveness of the signal decomposition-based approach [9, 10].

As the connection between a hazy component and its equivalent haze-free component, we describe the single-image dehazing problem as follows. Creating a mapping function that is impacted by the loss function L found in Eq. (6) is our goal in addressing this problem.

$$L(\emptyset) = \frac{1}{N} \sum_{j=1}^N \|F(Y_j, \emptyset) - X_j\|^2 \quad (6)$$

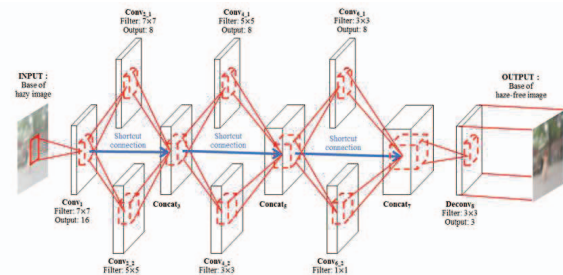


Fig. 4. Proposed Deep Convolutional Neural Network.

The set of parameters trained for DNN is represented by the symbol $\emptyset$. N is the number of possible combinations of training patches. The mapping function is denoted by F. Y_j and its corresponding patch X_j were taken from separate base components; one of them had a haze and the other did not.

B. Learning of DNN for Haze removal

Figure 4 shows how to use the recommended DNN to convert an input hazy base image element to its matching

haze-free component. The design comprises 8 layers. The initial layer is named Conv1 and utilizes 16 filters with a kernel size of 7×7 for convolution. The last layer consists of a deconvolution layer with three 3×3 kernel-sized filters. Additionally, the second, fourth, and sixth layers consist of convolutional layers on many scales with filter sizes of 7×7 , 5×5 , 5×5 , 3×3 , 3×3 , and 1×1 , in that order. Each sublayer is denoted by $Con_{m,n}$ n-th sub-layer. The upper portion of the filters produced 8 feature maps and the lower portions of the filters produced 16 feature maps. The ReLU function is used as an activation function for all convolution operations except the last layer. Although ReLU is primarily used for nonlinearity, it has been demonstrated that it can train more quickly than other non-linear separators.

Additionally, we suggest preserving the feature maps generated by the preceding layer and passing them on to the next layer to avoid color distortion in a dehazed image. To achieve this, we make use of three concatenation layers and three shortcut connections.

$$y = \sigma(x * W_{m,1}) + \sigma(x * W_{m,2}) + x \quad (7)$$

“*” indicates convolution, $W_{m,1}$, $W_{m,2}$ are filter weights and σ denotes the ReLU activation function. The three objects that a concatenation layer produces are combined using the “+” operation. Feature maps of different depths created by the three elements are combined using the concatenation layer rather than element-wise addition as in [11], before passing them onto the subsequent layer. CNN can utilize deep residual learning to get a dehazed base image component with little color distortion. Our CNN benefits from deep residual learning advantages, such as addressing the vanishing gradient problem and reducing gradient explosions. Additionally, our convolutional layers with multiple sizes of filters to extract information from multiscale images, aiding in the restoration of blurry images.

C. Reconstruction of Haze-Free image

By adding the dehazed base part and enhanced detail part we will get the dehazed (haze-removed) image J.

$$J = Detail_I_{Base\ part} + Enhanced_I_{Detail\ part} \quad (8)$$

$Enhanced_I_{Detail\ part} = E \times I_{Detail\ part}$. where E is a parameter associated with the entropy of the input image I, which is discovered by nonlinearly regressing 30,000 pairs of training images to examine the connection between each pair of detail images and their ground truth.

A distinctive feature of our technique is that the performance of the suggested dehazing procedure will be steadily improved by executing iteratively. In other words, a better dehazing outcome would be obtained by re-inputting a hazy image into the system after it has been output haze removed. This benefit is mostly attributable to the fact that, for each dehazing operation, the input hazy picture is first divided into the base and the detail components. According to our CNN, even if the dehazing procedure is carried out again in an iterative fashion, the picture details would be well kept and the color distortion would be prevented.

IV. EXPERIMENTAL RESULTS

The Outdoor Dataset-OTS consists of 313,950 synthetic images, each of which is composed of a hazy image and its haze-free ground truth. These images were created using real-world outdoor sceneries that were gathered. We created our training dataset by randomly selecting numerous pairs of 16 by 16 patches from each hazy image and the corresponding haze-free ground truth in OTS. We were able to collect a total of 1,000,000 pairs of patches as a consequence. We used soft matting image filtering as part of our preprocessing method to obtain the training patch's fundamental component for each image patch [8]. The initialization of the filter weights occurs during training by selecting random numbers from a Gaussian distribution. We train the network across 100 epochs with a batch size of 20 with the learning rate set at 1×10^{-5} .

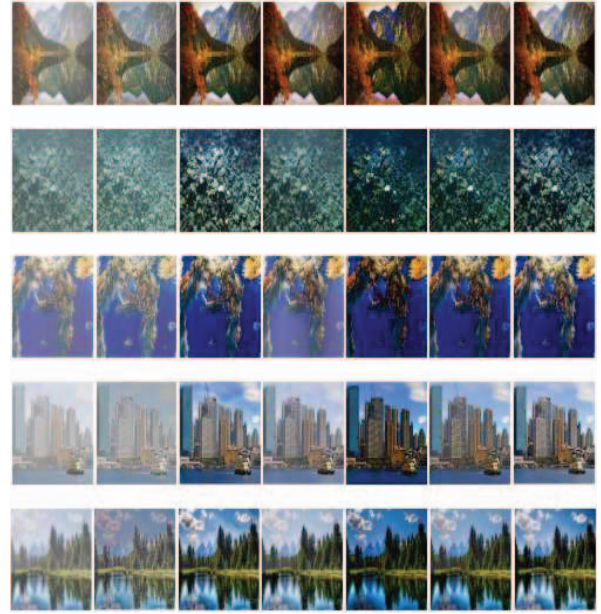


Fig. 5. Comparison of hazy images and the dehazed output images by proposed method and existing works (a) Haze image (b) He et al. [14] (c) Tarel et al. [12] (d) Cai et al. [13] (e) Zhu et al. [15] (f) Proposed work (g) Ground truth image.



Fig. 6. Comparison of the dehazed synthetic indoor images by proposed method and existing works (a) Haze image (b) He et al. [14] (c) Tarel et al. [12] (d) Cai et al. [13] (e) Zhu et al. [15] (f) Proposed work (g) Ground truth image.

Results are more accurate for the proposed ACNN-based image detection mode when compared with the existing network models. The satisfactory loss function value also is attained by the proposed model. In this model, the accuracy and error functions are good concerning the no. of iterations. For more iterations, the accuracy is increased, and the error is decreased.

The proposed dehazing model uses outdoor images as input and evaluates its effectiveness using ground truth images. The proposed dehazing model output images are compared with the existing dehazing model output images for qualitative comparison, as depicted in Figure 5.

In Figure 6, the dehazed synthetic indoor images are compared with respect to the other works. The assessment metrics, including PSNR and SSIM values [22, 23], of this study are compared with those of previous works in Table 1 and Table 2, respectively. For comparison of the results, six sample outdoor images are considered and applied to the proposed dehazing model. PSNR and SSIM values are evaluated for the outcomes and the average of PSNR and SSIM is calculated for comparative analysis.

TABLE I.
COMPARISON TABLE OF PSNR VALUES OF THE DEHAZING WORKS

Test Images	PSNR				
Outdoor Images	Tarel et al. [12]	He et al. [14]	Cai et al. [13]	Zhu et al. [15]	Proposed
Cityscape	13.58	22.74	19.88	20.65	24.69
Landscape1	14.02	26.47	25.35	15.29	26.58
Landscape2	14.63	23.6	18.02	17.23	24.98
Crab	12.1	14.25	12.99	15.29	22.97
Corel-reef	11.74	19.78	18.07	16.25	23.11
Average	13.214	21.368	18.862	16.942	24.466

The average PSNR value obtained by the proposed work is improved by 85.15%, 14.5%, 29.7%, and 44.4% when compared to the dehazing works of Tarel et al. [12], He et al. [14], Cai et al. [13], and Zhu et al. [15] respectively.

Similarly, the average SSIM of the proposed work is increased by 32%, 1.39%, 10.6%, and 8.1% to the dehazing model provided by Tarel et al. [12], He et al. [14], Cai et al. [13], and Zhu et al. [15] respectively. Figure 6 shows the PSNR value comparison with other works, while Figure 7 shows the SSIM value comparison with other works.

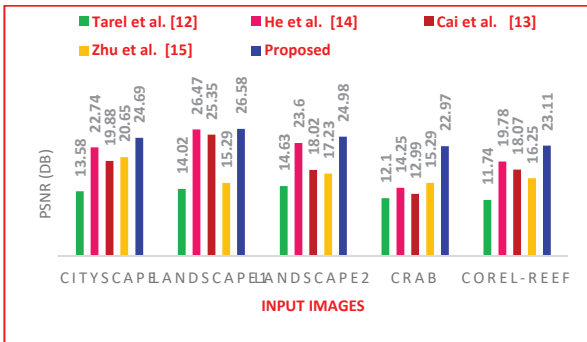


Fig. 6. PSNR of the proposed work comparison with other works.

TABLE II
SSIM OF THE PROPOSED TECHNIQUE COMPARISON WITH EXISTING TECHNIQUES

Test Images	SSIM				
Outdoor image	Tarel et al. [12]	He et al. [14]	Cai et al. [13]	Zhu et al. [15]	Proposed
Cityscape	0.774	0.941	0.904	0.859	0.958
Landscape1	0.667	0.936	0.937	0.882	0.945
Landscape2	0.742	0.92	0.848	0.885	0.938
Crab	0.702	0.936	0.743	0.863	0.945
Corel-reef	0.691	0.925	0.837	0.877	0.937
Average	0.7152	0.9316	0.8538	0.8732	0.9446

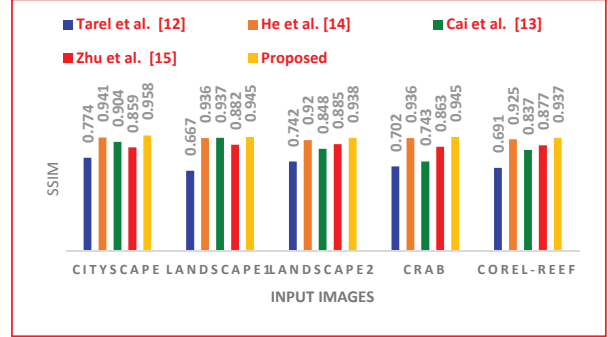


Fig 7. SSIM values comparison between proposed and other works.

V. CONCLUSION

For computer vision applications, the haze needs to be removed using an efficient dehazing method. In this work, an improved image dehazing method is developed and implemented using Deep Neural Networks. Convolution-based networks are used to optimize the dehazing process at various levels of filtering. The proposed dehazing technique using deep learning concepts is given to the outdoor datasets to analyze the metrics. The PSNR and SSIM are evaluated for the outcomes produced by the suggested work and distinguished with the state of the artworks. The quantitative improved values of the metrics are presented in this paper.

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